landau: an open source library for thermodynamic equilibria

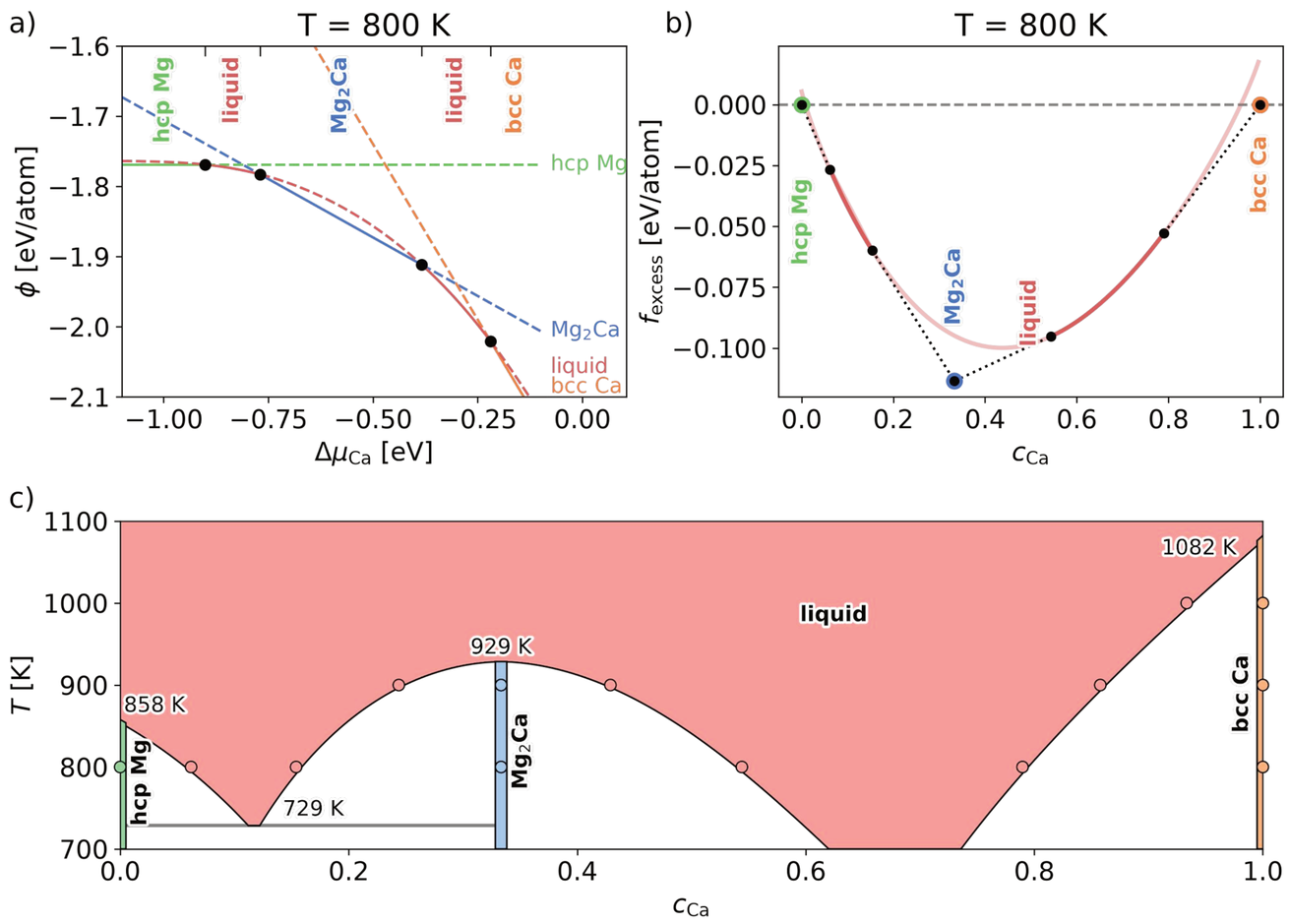
M. Poul (*marvinpo@umich.edu*), Thomas D. Swinburne (tswin@umich.edu), J. Neugebauer (neugebauer@mpi-susmat.de)

UNIVERSITY OF MICHIGAN

MAX PLANCK INSTITUTE FOR SUSTAINABLE MATERIALS

Motivation

Modern Machine Learning Potentials are increasingly accurate enough to enable quantitative prediction of alloy thermodynamics and phase diagrams. However, atomistic methods for free energy calculations and their output are not always suited to substitute directly into CALPHAD formalism. *landau* is written to accommodate atomistic free energy methods and produce publication ready figures.



Basics

- transitions calculated in semi-grand ensemble
- transitions are intersections (panel a)
- equivalent to common tangent (panel b)
- done over temperature range constructs the entire diagram (panel c)

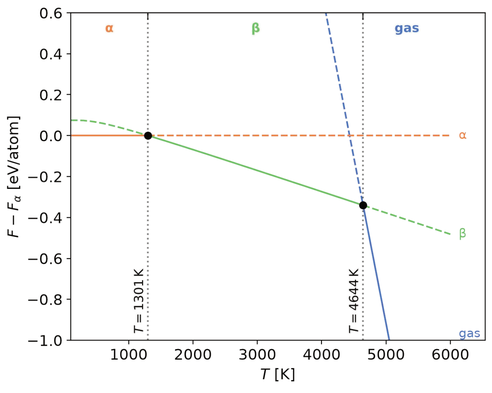
Software

github.com/eisenforschung/landau

Documentation

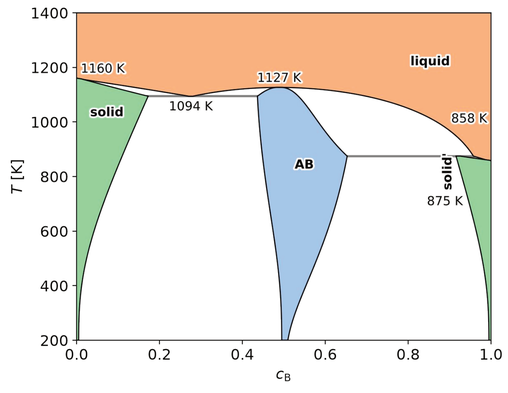
landau.rtfd.io

Example Diagrams by *landau*



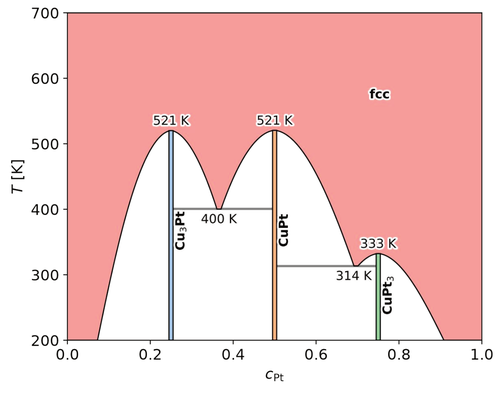
1D & 2D phase diagrams in T/mu/c

find and label transitions



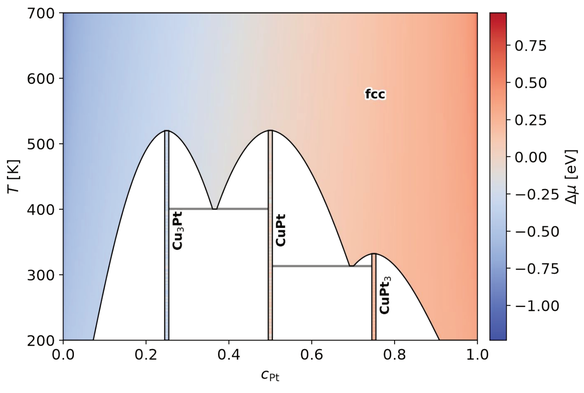
includes common phase models, e.g. point defects (above) or solutions (below)

smart, automatic label placement

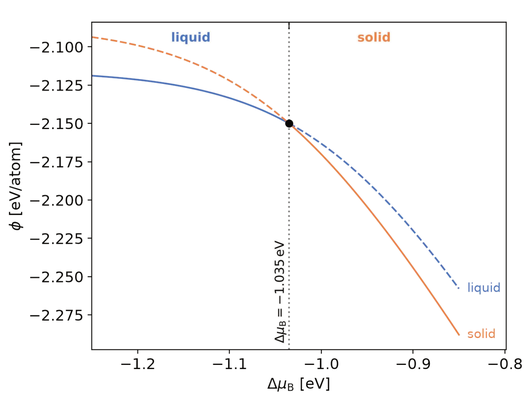


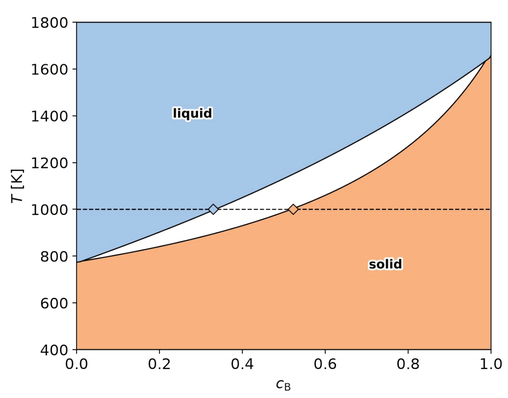
natively imports objects from phonopy or ASE

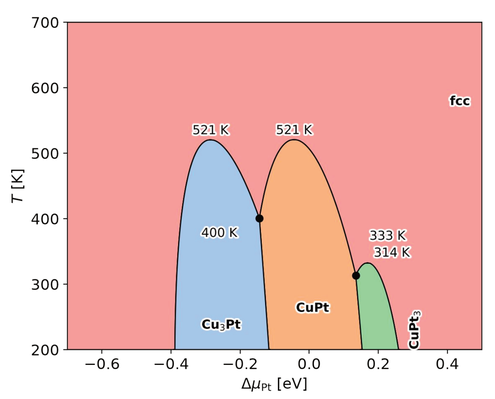
equivalent views in T-c or T-μ space

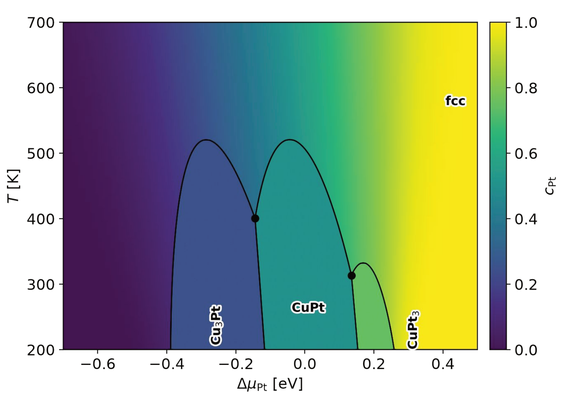


phase fields can be colored by additional features









References & Resources

Code for all examples directly by *landau*

github.com/pmrv/landau-gallery-poster

Menon et al., arXiv:2607.14795 (2026)

Poul et al., npj Comput. Mater. 11, 174 (2025)

Real World Examples

